Structural dependence of firms' growth in inter-firm trading network

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Abstract

Growth rates of sales of firms are known to follow a long-tailed distribution. This empirical law is explained by the generalized central limit theorem under the assumption of firms' independent stochastic growth, however, the other study insists that firms' network geometry is playing an essential role. In this paper, we analyze firms' growth based on big data of firms' profiles paying attention to the inter-firm network structure. We conclude that growth rates between pairs with direct transaction are significantly correlated, and therefore the network structure cannot be ignored in the process of firm's growth.

Keyword: Econophysics, complex network, Fisher's exact test

We analyze 3.8 million transaction data by one million Japanese firms and the firms' profile data for the last 20 years. As a focus of our study, we observe the synchronization of growth between firms linked by a trade-relationship.

Reference [1] showed that the growth rate of a business firm, which is defined as a year-over-year growth rates in sales, obeys a fat-tailed distribution such as a stable distribution confirmed from the data, and it is discussed that such a property can be understood theoretically by assuming an independent random multiplicative growth. On the other hand, Tamura et al. reported existence of a gravity-type interaction among firms indicating the important role of the geometry of transaction network structure [2]. In this study, we analyze the effect of the inter-firm network structure to the growth of adjoining firms (Fig.1).

We check the correlation of the growth rates between adjoining firms in the network by using Fisher's exact test. By comparing the result with randomly reconnected artificial networks, the significant difference is detected. This result suggests that the network structure cannot be ignored in the process of firm's growth.

Further, we search for characteristic network subgraphs which bring strong positive or negative correlation in growth rate.

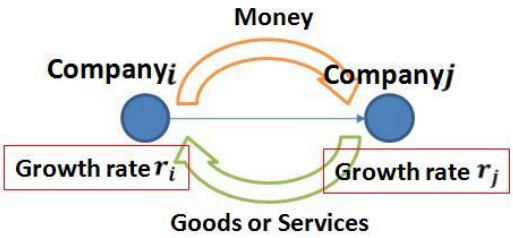


Figure 1 : Growth rates between adjoining firms in inter-firm trading network

References

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